

Shreya Pramanik

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Education

University of Minnesota

M.S. Statistics — GPA: 3.5/4.0

Coursework: Data Mining, Signal Processing, Regression Analysis, Advanced Probability, Bayesian Analysis

Sep 2023 – Aug 2026

Minneapolis, MN

Indian Institute of Technology Kanpur

M.S. Statistics — GPA: 3.8/4.0

Coursework: Time Series Analysis, Statistical Simulation, Design of Experiments, Multivariate Analysis

Sep 2020 – Jun 2022

Kanpur, India

Presidency University

B.S. Statistics — GPA: 3.7/4.0

Jul 2017 – Jul 2020

Kolkata, India

Technical Skills

Languages & Frameworks: Python, R, SQL, NoSQL, JavaScript, PyTorch, TensorFlow, LangChain, LangSmith, Streamlit, Power BI

DevOps Tools: Git, GitHub, Docker, RESTful APIs, FastAPI, Kubernetes, AWS (S3, EC2, EKS), CI/CD

Miscellaneous: Object-Oriented Programming, Machine Learning, Flow Matching Models, A/B Testing, RAG Pipelines, ChromaDB

Experience

HSBC Electronic Data Processing Pvt. Ltd

Jul 2022 – Jul 2023

Data Analyst, Wealth and Personal Banking

Bengaluru, India

- Drove **80–86% forecast accuracy** across **5+ APAC contact centers** by deploying an ensemble of **XGBoost**, **LSTM**, and **Prophet** demand models, driving real-time staffing for hundreds of daily customer interactions.
- Cut manual effort on weekly KPI packs by querying large-scale customer-contact data in **SQL** and automating **Power BI** summaries alongside Excel pivot-table and lookup-driven reporting tools.
- Detected demand surges during **Malaysia digital-banking rollouts** by shipping a production-ready modeling pipeline that informed regional **workforce-planning decisions**.

University of Minnesota: School of Statistics and Carlson School of Management

Sep 2023 – May 2026

Graduate Teaching Assistant; Graduate Research Assistant

Minneapolis, MN

- Mentored **100+ graduate students** to working proficiency in **R** by leading coding labs for regression and simulation courses.
- Delivered analysis-ready data for a longitudinal reduced-work-week policy study by cleaning **654 survey responses** (flagged ~5% low-quality) and coding **16 interview transcripts** at **Cohen's $\kappa = 0.78$** .

Projects

Flight Passenger Rights RAG Assistant | *FastAPI, PostgreSQL, Docker, AWS EKS* [🔗](#) [🐙](#)

Sep 2026

- Built a **passenger-rights RAG assistant** for passengers with delayed or cancelled flights, covering **EU261**, **UK261**, and **U.S. DOT** rules across four U.S. airlines served through **FastAPI**; achieved **74% complete evidence retrieval**, evidence recall 0.57, MRR 0.65 using **PostgreSQL** with pgvector, full-text search, and BGE cross-encoder reranking.
- Prevented wrong-jurisdiction answers that could misstate up to \$600 in passenger compensation by routing governing law from the departure airport and validating every cited sentence, achieving **100% governing-law coverage**, **0.70 faithfulness**, and **zero fabricated citations** on gpt-oss-20b; independently evaluated by a **LLM judge**.
- Deployed on **AWS EKS** behind an NLB with two API replicas, readiness gates, and a PodDisruptionBudget for safer rollouts; automated provisioning with **Terraform**, **eksctl**, and **Helm**, protected live PostgreSQL data on encrypted **EBS**, **restricted S3 backup** writes through **IRSA**, and monitored failures with **Grafana** and SNS alerts.

E-Commerce Analytics Platform | *MySQL, Python, Power BI, DAX, GitHub Actions* [🔗](#) [🐙](#)

Jun 2026 – Jul 2026

- Built an end-to-end **ETL pipeline** in Python (pandas, SQLAlchemy) to transform **100K+ raw transactions** into a queryable MySQL star schema, with structured staging, normalization, and database-level integrity constraints.
- Developed a **Power BI dashboard** using SQL and DAX to analyze **\$13.17M in revenue**, isolating checkout as the top drop-off stage (**11.56%**), organic search as **62x more efficient** than paid, and VIP customers spending **2.3x** more.
- Engineered a reproducible **A/B testing framework** with deterministic variant assignment and SQL-based statistical testing, identifying a **significant 31% conversion lift**, automated testing and SQL linting through **GitHub Actions CI/CD**.

Financial Shock Effects on Hospital Operations | *Python, XGBoost, SHAP* [🔗](#) [🐙](#)

Jan 2026 – Feb 2026

- Predicted operating margins across **6,483 hospitals** by engineering **15+ features** from **66,679 HCRIS** cost reports (Medicaid share, uncompensated care, occupancy, staffing) and training **XGBoost** with hospital-level grouped cross-validation, holding **0.084 MAE** and **0.129 RMSE** on held-out data; **SHAP** surfaced occupancy and Medicaid exposure as dominant drivers.
- Quantified shock absorption under simulated **5–15% Medicaid reimbursement cuts**, showing low-exposure urban private hospitals (<20% Medicaid) lost under **1.2% of operating margin** while high-exposure urban teaching hospitals (40–60% Medicaid) turned financially fragile.
- Isolated the **top and bottom 20** most shock-absorptive institutions (**2011–2022**) for policy review by ranking every hospital on a composite resilience metric (Δ margin, post- vs. pre-shock).

Harmful Brain Activity Classification | *EfficientNet, TensorFlow, scikit-learn* [🔗](#) [🐙](#)

Feb 2025 – May 2025

- Reduced misclassification of **6 harmful brain activity types** by **39%** (KL divergence $0.85 \rightarrow 0.52$) by engineering a **dual-source 8-channel spectrogram pipeline** that fused Kaggle-provided spectrograms with EEG-derived mel spectrograms across **17,089 patient recordings** (HMS Kaggle competition).
- Held a best cross-validated **KL divergence of 0.52** across **5-fold GroupKFold** (grouped by patient ID to prevent leakage) by training **EfficientNetB0** on 512×512 spectrograms derived from **19 electrode-pair signals** across 4 brain regions (LL, RL, LP, RP), with cosine LR annealing and mixed-precision inference on dual T4 GPUs.

Achievements

- **Won HeatMap Hackathon 2026** (BData Inc. × American Burn Association × MN Healthcare) with a geospatial analysis translating burn injury data into equitable care access.
- Received the **Academic Excellence Award** for outstanding performance in M.S. Statistics, IIT Kanpur.